fancy-uc3m-thesis

Template Documentation — v1.0.1

Universidad Carlos III de Madrid thesis template

A Typst template for bachelor/master theses at Universidad Carlos III de Madrid, following university guidelines.

The template is based on ldcas-uc3m/thesis-template and clean-uc3m (a fork of clean-dhbw).

1. Installation

The recommended way to install it is through Typst Universe. You can initialize a new project from the template with:

```shell
typst init @preview/fancy-uc3m-thesis my-final-thesis
```

This creates a `my-final-thesis/` directory with all the files needed to get started.

1.1. Manual installation

Clone the repository, install [Just](https://github.com/casey/just) and run:

```
just install
```

Now you can initialize the template with:

```
typst init @local/fancy-uc3m-thesis my-final-thesis
```

2. Configuration

The template exports a single `conf` function, which you apply to your document with `#show: conf.with(...)`.

2.1. Minimal example

```typ
#import "@preview/fancy-uc3m-thesis:0.4.0": conf

#show: conf.with(
  title: "My Bachelor Thesis",
  author: "GUL UC3M",
  degree: "Computer Science and Engineering",
  advisors: ("Prof. L",),
  location: "Leganés, Madrid",
  thesis-type: "TFG",
  date: datetime(year: 2025, month: 6, day: 15),
  language: "en",
  format: "ieee",
  abstract: (
    body: [A short description of my thesis.],
```

```
    keywords: ("Keyword 1", "Keyword 2", "Keyword 3"),
  ),
  genai-declaration: (usage: false),
)

= Introduction

My thesis starts here.
```

2.2. Abstract

The `abstract` parameter takes a dictionary with the following keys:

Key	Type	Description
`body`*	`content`	The abstract text.
`keywords`*	`array` of `str`	Between 2 and 5 keywords. For IEE, see IEEE Taxonomy.

```typ
  abstract: (
  body: [
    This thesis presents an implementation of interrupts and timers
    in the CREATOR simulator.
  ],
  keywords: ("Operating Systems", "Interrupts", "RISC-V", "Simulation"),
),
```

For Spanish documents, it is required to provide an `english-abstract` with the same structure:

```typ
language: "es",
abstract: (
  body: [Una breve descripción de mi tesis.],
  keywords: ("Sistemas Operativos", "Interrupciones", "RISC-V"),
),
english-abstract: (
  body: [A brief description of my thesis.],
  keywords: ("Operating Systems", "Interrupts", "RISC-V"),
),
```

2.3. Epigraph

An optional opening quote placed before the abstract. Provide a dictionary with:

Key	Type	Description
`quote`*	`content`	The quote text.
`author`*	`str`	The author of the quote.
`source`	`str`	(Optional) The source where the quote was found.

```typ
epigraph: (
  quote: [The purpose of abstracting is not to be vague, but to create a new
  semantic level in which one can be absolutely precise.],
```

```
  author: "Edsger W. Dijkstra",
  source: "The Humble Programmer, 1972",
),
```

2.4. Outlines

The `outlines` parameter accepts an optional dictionary with the following boolean keys. All keys are optional (default `false`):

Key	Type	Description
`figures`	`bool`	Include a list of figures.
`tables`	`bool`	Include a list of tables.
`listings`	`bool`	Include a list of code listings.
`custom`	`array of content`	Additional custom outlines (e.g. algorithms).

```typ
outlines: (
  figures: true,
  tables: true,
  listings: false,
  // custom: (
  //   outline(
  //     title: [List of Algorithms],
  //     target: figure.where(kind: "algorithm"),
  //   ),
  // ),
),
```

2.5. Abbreviations

You can pass abbreviations as a dictionary mapping abbreviations to their expansions (automatically sorted alphabetically):

```typ
abbreviations: (
  API: "Application Programming Interface",
  CPU: "Central Processing Unit",
  ML: "Machine Learning",
  TFG: "Trabajo de Fin de Grado",
),
```

Or as custom `content` if you need full control over the layout:

```typ
abbreviations: [
  #table(
    columns: (auto, 1fr),
    [*API*], [Application Programming Interface],
    [*ML*],  [Machine Learning],
  )
],
```

2.6. Glossary

The template integrates with `glossarium` for automatic glossary and acronym management.

Define your entries in a separate file (e.g. `config/glossary.typ`):

```typ
#let glossary-entries = (
  (
    key: "API",
    short: "API",
    long: "Application Programming Interface",
    description: "A set of protocols and tools for building software.",
  ),
  (
    key: "ML",
    short: "ML",
    long: "Machine Learning",
    description: "A subset of AI that enables computers to learn from data.",
  ),
)
```

Then import it in your main file and pass it to `conf`:

```typ
#import "config/glossary.typ": glossary-entries

#show: conf.with(
  // ...
  glossary: glossary-entries,
)
```

To reference a glossary term in the body text, use `#gls("key")` (singular) or `#glspl("key")` (plural). These are re-exported from `glossarium` and available after the `#show: conf.with(...)` call:

```typ
#import "@preview/fancy-uc3m-thesis:0.4.0": conf
#import "@preview/glossarium:0.5.9": gls, glspl

// In the body:
The #gls("API") exposes several endpoints.
Multiple #glspl("ML") models were evaluated.
```

Alternatively, you can pass fully custom `content` to `glossary` if you don't want to use `glossarium`.

2.7. Generative AI Declaration

The `genai-declaration` parameter is **required** by the university. You can either use the built-in template or provide custom content.

2.7.1. Using the built-in template (recommended)

If you did not use generative AI (`usage: false`):

```typ
genai-declaration: (usage: false),
```

If you did use generative AI (`usage: true`), it is required to provide the full declaration:

Pass a dictionary with the following keys:

Key	Type	Required	Description
`usage`*	`bool`	Always	Whether you used generative AI in the thesis.
`data-usage`	`dictionary`	If `usage` is `true`	Data usage declaration. See below.
`technical-usage`	`dictionary`	If `usage` is `true`	Technical usage declaration. See below.
`usage-reflection`	`content`	If `usage` is `true`	Personal reflection on AI usage.

`data-usage` keys — all are `bool` (`true` = YES, `false` = NO):

Key	Description
`confidential`	Did you submit confidential data (trade secrets, proprietary info)?
`copyright`	Did you submit copyrighted materials?
`personal`	Did you submit personal data?
`tos`	Did your usage respect the tool's terms of use?

`technical-usage` keys — include only those that apply, with a brief description of the prompt and interaction. Items are automatically grouped into two sections matching the official form:

Documentation and drafting: `reflection`, `review`, `research`, `references`, `summary`, `translation`

Develop specific content: `assistance-coding`, `generating-content`, `optimization`, `data-processing`, `idea-inspiration`, `other`

An optional `tool` key (`str`) declares the AI system name and version (e.g. `"ChatGPT 4o"`).

```typ
genai-declaration: (
  usage: true,
  data-usage: (
    // true = YES / used with authorization; false = NO / not used
    confidential: false,
    copyright: false,
    personal: false,
    tos: true,
  ),
  technical-usage: (
    // Name of the AI system and version (optional):
    // tool: "ChatGPT 4o",

    // ── Documentation and drafting ──────────────────────────────
    reflection: [I asked for a list of alternatives to address the
      problem stated at the beginning of the paper.],
```

```
    // review: [...],
    // research: [...],
    // references: [...],
    // summary: [...],
    // translation: [...],

    // ── Develop specific content ──────────────────────────────────
    assistance-coding: [I asked for explanations about a Python code
      snippet and for help correcting an error.],
    // generating-content: [...],
    // optimization: [...],
    // data-processing: [...],
    // idea-inspiration: [...],
    // other: [...],
  ),
  usage-reflection: [
    Overall, I found generative AI useful for boilerplate tasks and
    quick explanations, but it required careful verification of all
    generated content. It did not replace critical thinking or
    domain-specific knowledge.
  ],
),
```

2.7.2. Providing custom content

If you need full control, pass any `content` directly:

```typ
genai-declaration: [
  // Your custom AI declaration content here.
],
```

3. Styles

The template offers three visual styles, controlled by the `style` parameter.

3.1. "fancy" (default)

An opinionated style inspired by the original LaTeX template, with:

- Decorative chapter title pages with a horizontal rule and chapter number/name centered
- Serif headers showing the current chapter name and page number
- APA or IEEE figure captions and table captions (depending on `format`)
- Libertinus Serif body font

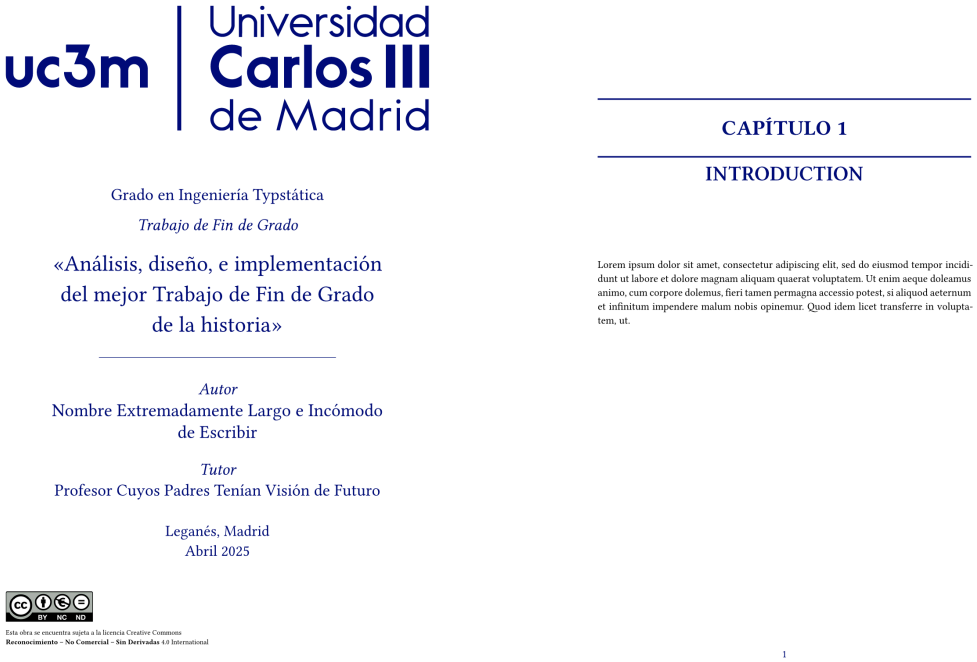


Figure 1: Example of the **fancy** style.

3.2. "clean"

A clean, minimal style inspired by clean-dhbw, with:

- Large chapter number in the background on chapter pages
- Simple running header showing the chapter name
- Clean-style figure and table captions
- Libertinus Serif body font

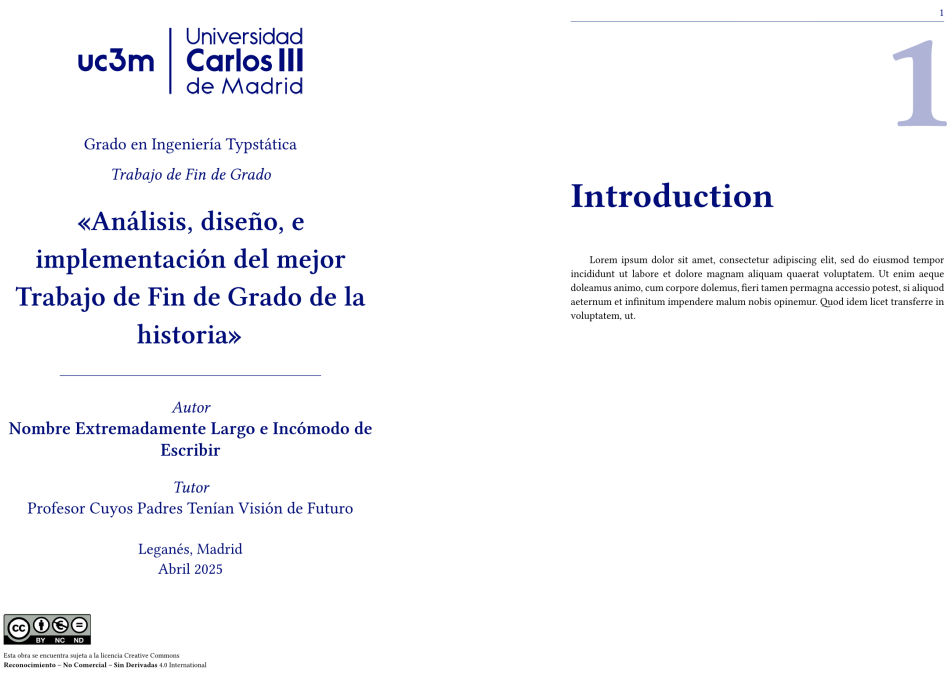


Figure 2: Example of the **clean** style.

3.3. `"strict"`

Strictly follows the UC3M library guidelines:

- Centered uppercase chapter headings
- Page numbers in the footer (Roman numerals in front matter, Arabic in body)
- APA or IEEE figure captions and table captions (depending on `format`)
- No decorative elements
- Double-sided layout is **not** permitted with this style
- Times New Roman font, 12pt

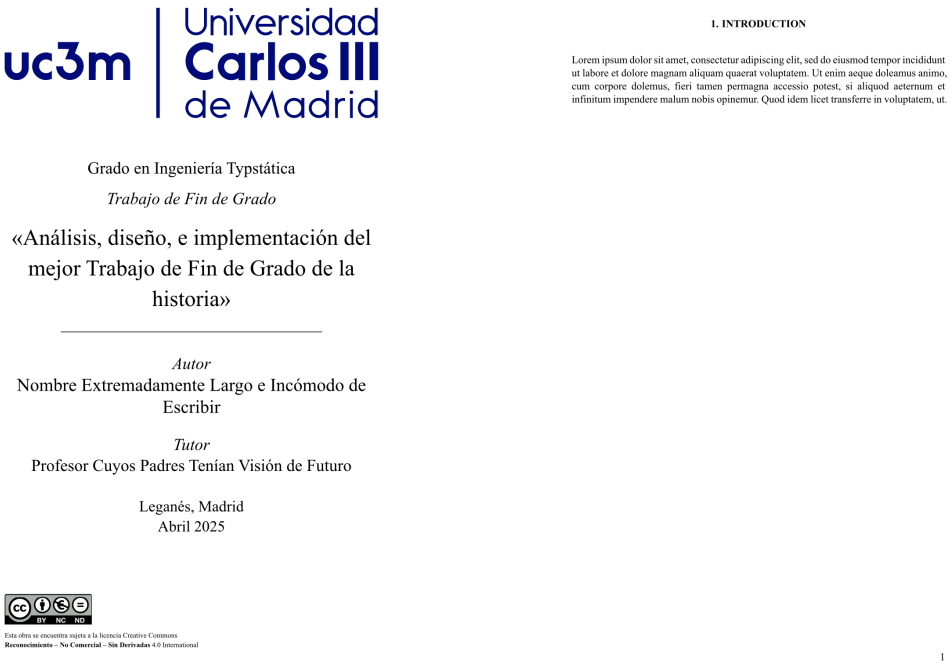


Figure 3: Example of the **strict** style.

> **Idea**
>
> You can mix styles for the title page and body independently using `titlepage-style`. For example, to use a `fancy` title page with a `strict` body: `style: "strict", titlepage-style: "fancy"`.

4. Table/Figure styles

Tables and figures also have different visual styles. These are automatically set depending on the `style` and `format` values, but can be overwritten through `table-style` and `figure-style`.

Both `"apa"` and `"ieee"` follow the university guidelines for APA/IEEE format, while the `"clean"` style is more cohesive.

Figure 8.1
El mejor logo de la UC3M, con diferencia



Table 8.1
Comparison of desktop Operating Systems

OS	Silksong	Ads	Spyware	Unix	FOSS	Penguins
Linux	✓			✓	✓	✓
MacOS	✓		✓	✓		
Windows	✓	✓	✓			

Figure 4: Example of the **"apa"** figure style.



Fig. 8.1. El mejor logo de la UC3M, con diferencia

Table 8.1
Comparison of desktop Operating Systems

OS	Silksong	Ads	Spyware	Unix	FOSS	Penguins
Linux	✓			✓	✓	✓
MacOS	✓		✓	✓		
Windows	✓	✓	✓			

Figure 5: Example of the **"ieee"** figure style.



Figure 8.1: El mejor logo de la UC3M, con diferencia

Table 8.1: Comparison of desktop Operating Systems

OS	Silksong	Ads	Spyware	Unix	FOSS	Penguins
Linux	✓			✓	✓	✓
MacOS	✓		✓	✓		
Windows	✓	✓	✓			

Figure 6: Example of the **`"clean"`** figure style.

Changelog

v1.0.1

- Improve title rendering in header for long titles

API Reference

conf

Main configuration function.

Recommended to use with `#show: conf.with(...)`.

Parameters

```
conf(
  doc: content,
  title: str,
  author: str,
  degree: str,
  advisors: array,
  location: str,
  thesis-type: str,
  date: datetime,
  bibliography-content: content,
  language: str,
  format: str,
  style: str,
  titlepage-style: str auto,
  table-style: str auto,
  figure-style: str auto,
  figure-spacing: length none,
  font: string auto,
  font-titlepage-size: lenght,
  double-sided: bool,
  logo: str,
  short-title: str,
  date-format: str auto,
  license: bool,
  flyleaf: bool,
  epigraph: dictionary none,
  abstract: dictionary,
  english-abstract: dictionary,
  acknowledgements: content none,
  outlines: dictionaty none,
  appendixes: content none,
  glossary: array content none,
  abbreviations: dictionary content none,
  genai-declaration: dictionary content
) -> content
```

doc `content`

Thesis contents.

title `str`

Title of the thesis.

Default: none

author `str`

Author name.

Default: none

degree `str`

Degree name (e.g. `"Computer Science and Engineering"`).

Default: none

advisors `array`

List of advisor names.

Default: none

location `str`

Presentation location.

Default: none

thesis-type `str`

Type of thesis (`"TFG"` or `"TFM"`).

Default: none

date `datetime`

Presentation date.

Default: none

bibliography-content `content`

Bibliography contents, usually obtained by calling `bibliography`.

Default: none

language `str`

`"en"` or `"es"`.

Default: `none`

format `str`

Thesis format, either `"apa"` or `"ieee"`.

Default: `none`

style `str`

Visual style, mainly affecting headings, headers, and footers. The available styles are `strict`, which strictly follow's the university library's guidelines, `clean`, based on clean-dhbw, and `fancy`, based on my original LaTeX version.

Default: `"fancy"`

titlepage-style `str` or `auto`

Style for the titlepage (see `style`). If set to `auto`, uses the main style.

Default: `auto`

table-style `str` or `auto`

Style for the table caption, either `"clean"`, `"apa"`, or `"ieee"`. If set to `auto`, uses the default for the main style (`"clean"` for `clean` style, `format` for the rest).

Default: `auto`

figure-style `str` or `auto`

Style for the figure caption, either `"clean"`, `"apa"`, or `"ieee"`. If set to `auto`, uses the default for the main style (`"clean"` for `clean` style, `format` for the rest).

Default: `auto`

figure-spacing `length` or `none`

Extra spacing to give to figures and tables. If `none`, no extra spacing.

Default: `0.75em`

font `string` or `auto`

Font to use. By default, `"Libertinus Serif"` in all styles except `"strict"`, where it's `"Times New Roman"`. Can't be set with the `"strict"` style.

Default: `auto`

font-titlepage-size `lenght`

Font size in the titlepage. Useful when messing with the `font` parameter.

Default: `16pt`

double-sided `bool`

Whether to use double-sided pages. This is not allowed in the `strict` style.

Default: `false`

logo `str`

Type of logo (`"old"` or `"new"`).

Default: `"new"`

short-title `str`

Shorter version of the title, to be displayed in the headers. Only applies if `double-sided` is set to `true`.

Default: `none`

date-format `str` or `auto`

Date format. Use `auto` or specify the format using the [Typst format syntax](https://typst.app/docs/reference/foundations/datetime/#format).

Default: `auto`

license `bool`

Whether to include the CC BY-NC-ND 4.0 license.

Default: `true`

flyleaf `bool`

Whether to include a blank page after the cover.

Default: `true`

epigraph `dictionary` or `none`

A short quote that guided you through the writting of the thesis, your degree, or your life. Consists of `quote` (of type `content`), the body or text itself, `author` (of type `str`), the author of the quote and, optionally, `source` (of type `str`), where the quote was found.

Default: `none`

abstract `dictionary`

A short and precise representation of the thesis content. Consists of `body` (of type `content`), the main text, and `keywords`, an array of key terms (of type `str`) (see [IEEE Taxonomy](https://www.ieee.org/content/dam/ieee-org/ieee/web/org/pubs/ieee-taxonomy.pdf)).

Default: `none`

english-abstract `dictionary`

An english translation of the abstract. Compulsory for spanish works, invalid for english ones.

Default: `none`

acknowledgements `content` or `none`

Text where you give thanks to everyone that helped you.

Default: `none`

outlines `dictionaty` or `none`

Set of extra outlines to include (`figures`, `tables`, `listings`), and extra custom outlines (`custom`, an array of `contents` – typically the result of calling `outline`).

Default: `none`

appendixes `content` or `none`

Set of appendixes.

Default: `none`

glossary `array` or `content` or `none`

Glossary entries. If `array` is provided, it will use the `glossarium` library. If content is passed, it will display that content, without applying any styling.

Default: `none`

abbreviations `dictionary` or `content` or `none`

Abbreviations, acronyms and initials used throughout the thesis. You can provide a map (dictionary of strings) or a custom one (`content`).

Default: `none`

genai-declaration `dictionary` or `content`

Information about the use of Generative AI in the thesis. You can suply your own `content`, or use the university's template, by suplying a `dictionary`. See the example for more details.

Default: `none`